

{ Layer 3 }

- Post-training teaches base models to follow instructions, formatting and act like conversational assistants



• Supervised fine-tuning

↳ Feeds the model thousands of high quality

"instruction response pairs" (prompt: "write a poem", response: [—])

↳ it teaches the model basic structure of conversations



↳ Limitation: SFT teaching by

Limitation

→ Learn to answer

→ No awareness of this answer is good or not

→ Small Data Set so it's likely will overfit

RLHF : Separate Reward model

Take User preferences answers → train the model to give $\oplus$ or $\ominus$ to the Response

ex ("Chat GPT 4 trained on") =

$$\text{Loss} = -\log(\text{Reward}_{\text{better}} - \text{Reward}_{\text{worse}})$$

after training the Reward model → we can use the output with algorithm like PPO, GRPO or any RL algo can be used in NN

So For the same Prompt :

Prompt
[Best Ai student in the world?]

not helpful → General $\ominus$
[Could be anyone]

good one xd $\oplus$
[Ahmad Al-Hrouz]

Reward { we can forward to the model
is this answer good or not

Cons: biasing → specific words
→ long answer
Since human will choose
; Reward Hacking, confident style over factual truth

DPO

: Direct Preference Optimization

✗ Skipping building separate model reward

→ instead, use math to adjust weights based

Chosen, Rejected Pair

Cons

: if the data are messy the model will die with the data

→ model quality will drop

RLVR and GPRO : Reinforcement Learning with verifiable Rewards

Human preferences are subjective, which is Bad for logic heavy tasks, **RLVR** uses objective, verifiable Reward

When to use : Math, coding $\rightarrow$ u don't need human to say if this code is good or Bad $\rightarrow$ Run the code
So if it's good or not, Math equation are balanced
... etc

IPo, Kto, ORPO, RLAI F

other algo

Safety tuning : Shapes Resful behavior, Policy adherence, harmful Request
Sensitive data handling

Like Deepsets about **tiwan** XD